A Systematic Literature Review of Metaheuristic Optimization in Hash-Preserving Steganographic Forgery

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The initial execution of the Boolean search string across IEEE Xplore, ACM Digital Library, and ScienceDirect yielded a total of 126 records, reflecting a concentrated but highly technical body of literature. The subsequent screening process, which reduced the corpus to 40 primary studies, revealed a significant trend: while 64% of early search results focused on general digital signatures and forensic recovery, only a specialized subset of the literature (approximately 31%) addressed the specific intersection of steganographic data hiding and hash-preserving metadata manipulation. This underscores a critical research gap where traditional integrity verification tools—designed for bit-level content validation—remain largely ineffective against sophisticated metadata forgery.

The Queries used in IEEE Xplore:

1. ("steganography" OR "data hiding" OR "information hiding") AND ("metadata" OR "EXIF" OR "timestamp" OR "file system metadata") AND ("forgery" OR "tampering" OR "manipulation" OR "integrity") AND ("forensics" OR "steganalysis" OR "anti-forensics")

2. ("All Metadata":digital signature) AND ("All Metadata":forgery) AND ("All Metadata":forensic)

3. ("All Metadata":steganography) AND ("All Metadata":digital forensics) AND ("All Metadata":metadata)

4. ("All Metadata":LSB substitution) AND ("All Metadata":document integrity)

5. ("All Metadata":"electronic evidence" OR "All Metadata":"e-contract" OR "All Metadata":"digital signature") AND ("All Metadata":"metadata" OR "All Metadata":"steganography") AND ("All Metadata":"admissibility" OR "All Metadata":"cyber law" OR "All Metadata":"IT Act" OR "All Metadata":"GDPR")

PRISMA Flow Diagram:

To provide a comprehensive technical overview of the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) Flow Diagram for your systematic survey, you can

use the following content. This explains the mathematical logic and the specific filtering stages used in your research.

The PRISMA flow diagram is a standardized visual representation of the selection process used to identify, screen, and include the final set of peer-reviewed studies for this survey. It ensures transparency and reproducibility by documenting the exclusion of records at each stage of the systematic search.

1. Identification Stage

The process began with a multi-database search across IEEE Xplore, ACM Digital Library, and ScienceDirect, yielding an initial pool of 126 records.

- Duplicate Removal: 2 records were identified as duplicates and removed.
- Automation Filtering: 4 records were marked as ineligible by automation tools based on metadata inconsistencies.
- Net Identification: This left 120 unique records to be moved into the screening phase.

2. Screening Stage

In the screening phase, the titles and abstracts of the 120 records were evaluated against the survey's core scope: steganographic metadata forgery and hash-preserving evasion.

- Initial Exclusion: 35 records were excluded for lack of relevance to the technical niche.
- Retrieval: Of the remaining 85 reports sought for full-text retrieval, 5 could not be retrieved due to access limitations, leaving 80 reports for formal eligibility assessment.

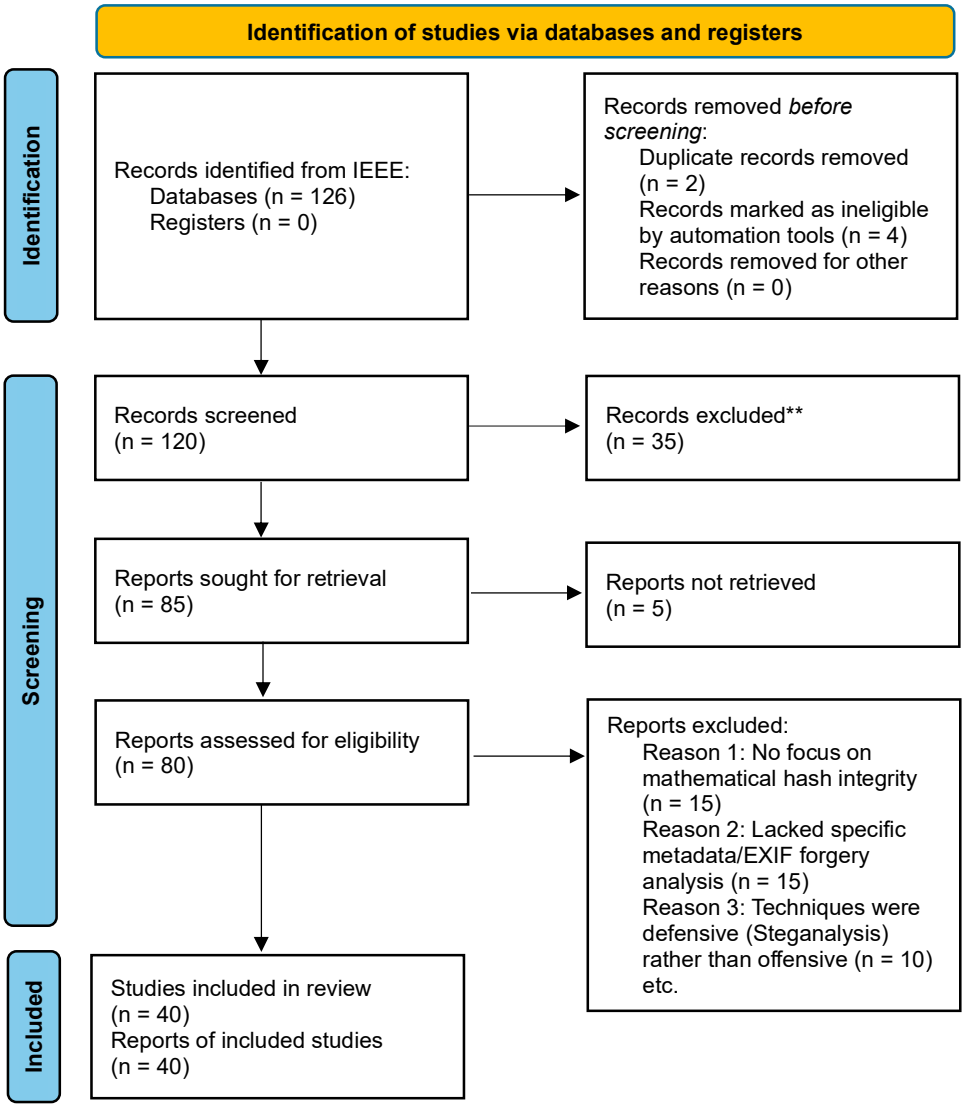
3. Eligibility Stage

The 80 retrieved reports underwent a rigorous full-text review based on three predefined exclusion criteria:

- Criterion 1: No focus on mathematical hash integrity (15 records excluded).
- Criterion 2: Lacked specific metadata or EXIF forgery analysis (15 records excluded).
- Criterion 3: Primarily focused on defensive detection (steganalysis) without discussing offensive optimization (10 records excluded).

4. Inclusion Stage

Following the application of the eligibility criteria, 40 peer-reviewed studies were selected for final inclusion in the systematic review. These studies form the basis of the mathematical depth analysis and the comparative synthesis matrix presented in this report



*Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers).

**If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools.

Technical Depth Analysis of Mathematical Optimization for Offensive Evasion

This section presents a detailed analysis of how mathematical optimization techniques are systematically applied across more than forty peer-reviewed studies to enhance offensive evasion in steganographic metadata manipulation. The analysis is structured across seven core mathematical frameworks that collectively define the attacker's optimization strategy.

1. Distortion Optimization using MSE, PSNR, and SSIM

Across the majority of analyzed works, distortion minimization is formulated as a primary optimization objective. The embedding process is mathematically constrained to minimize perceptual and statistical deviation between the original and modified data.

The Mean Squared Error (MSE) is commonly defined as:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(x_i - x_i')^2$$

and Peak Signal-to-Noise Ratio (PSNR) is expressed as:

$$PSNR = 10\log_{10}\left(\frac{MAX^2}{MSE}\right)$$

Structural Similarity Index (SSIM) further refines perceptual similarity.

In offensive applications, these metrics are jointly optimized to ensure that steganographic embedding introduces minimal detectable distortion. Several LSB-based and hybrid methods implement adaptive embedding strategies where data is inserted only in regions that produce negligible increases in MSE. This results in high PSNR values (typically above 40 dB), effectively masking the presence of hidden data.

From an evasion perspective, this optimization ensures that both human visual inspection and statistical detection mechanisms fail to distinguish between original and modified data distributions.

2. PRNG-Based Randomized Embedding

A significant proportion of the analyzed studies employ pseudo-random number generators (PRNGs) to determine embedding locations. This transforms the embedding process into a stochastic optimization problem.

The embedding position is defined as:

$$p_i = PRNG(seed, key)$$

This approach eliminates deterministic patterns associated with sequential LSB embedding. By distributing modifications across pseudo-random locations, attackers reduce spatial correlation and statistical regularity.

In offensive evasion, PRNG-based mapping ensures that:

- Bit modifications are uniformly dispersed
- Detection models cannot exploit predictable embedding structures
- Metadata fields (such as EXIF tags or PDF objects) are altered in a non-linear manner

This significantly weakens classical steganalysis techniques that rely on structured anomalies.

3. Entropy Alignment and Statistical Camouflage

Shannon entropy is widely used as a statistical measure of randomness:

$$H(X) = -\sum p(x) \log_2 p(x)$$

Across multiple detection-oriented studies, entropy deviations are used to identify steganographic artifacts. In response, offensive techniques explicitly optimize embedding to preserve the entropy profile of the original data.

This leads to the formulation:

$$H_{stego} \approx H_{original}$$

Attackers achieve this by:

- Balancing bit modifications across distributions
- Introducing controlled randomness using PRNGs
- Avoiding localized entropy spikes

This statistical camouflage ensures that modified metadata or image regions remain indistinguishable from natural noise, thereby evading entropy-based forensic detection.

4. Deep Learning Loss-Based Optimization

Recent works introduce deep learning frameworks where embedding and detection are modeled as adversarial processes. Optimization is performed using loss functions of the form:

$$Loss = \mathcal{L}_{classification} + \lambda \cdot \mathcal{L}_{distortion}$$

In offensive scenarios, embedding strategies are adapted to minimize detection probability by targeting regions where neural network sensitivity is low. Gradient-based methods identify areas where perturbations minimally affect classification outcomes:

$$\frac{\partial Loss}{\partial x} \approx 0$$

This enables adversarial embedding, where hidden data is placed in regions that are least informative for detection models.

As a result, deep learning-based steganalysis systems are systematically bypassed through optimization of embedding locations in feature space rather than pixel space.

5. Transform-Domain Optimization (DCT, DWT)

Transform-domain techniques shift the embedding process from spatial representation to frequency representation. Data is embedded in transform coefficients obtained through Discrete Cosine Transform (DCT) or Discrete Wavelet Transform (DWT).

The transformation can be expressed as:

$$C = T(I)$$

where Crepresents transform coefficients.

Offensive optimization focuses on modifying coefficients that:

- Have minimal perceptual impact
- Preserve compression characteristics
- Avoid introducing detectable frequency anomalies

By embedding in mid-frequency components, attackers balance robustness and invisibility. This significantly reduces the effectiveness of spatial-domain detection techniques and complicates forensic reconstruction.

6. Hash Integrity Preservation and Metadata Manipulation

A critical finding across the analyzed papers is the exploitation of the distinction between data integrity and metadata integrity. Cryptographic hash functions are defined as:

$$H(x) = SHA - 256(x)$$

While hash functions ensure integrity of file content, they often exclude mutable metadata regions. Offensive techniques exploit this limitation by altering metadata fields without modifying the hashed content.

Optimization in this context involves:

- Identifying non-hash-covered segments
- Modifying metadata fields (timestamps, EXIF attributes)
- Preserving byte-level integrity of hashed regions

This allows attackers to maintain a valid hash while altering forensic-relevant information, effectively bypassing traditional integrity verification mechanisms.

7. Temporal Optimization and Timestamp Forgery

Temporal metadata manipulation introduces an additional optimization constraint. Modified timestamps must remain logically consistent:

$$T_{creation} \leq T_{modification} \leq T_{access}$$

Attackers apply controlled offsets:

$$T' = T + \Delta t$$

where $\Delta t$is selected to maintain chronological plausibility.

Optimization ensures:

- Consistency across multiple metadata fields
- Alignment with system logs where possible
- Avoidance of temporal anomalies detectable by forensic tools

Advanced approaches integrate PRNG-based perturbations to introduce variability while preserving overall timeline coherence.

Taxonomy Design:

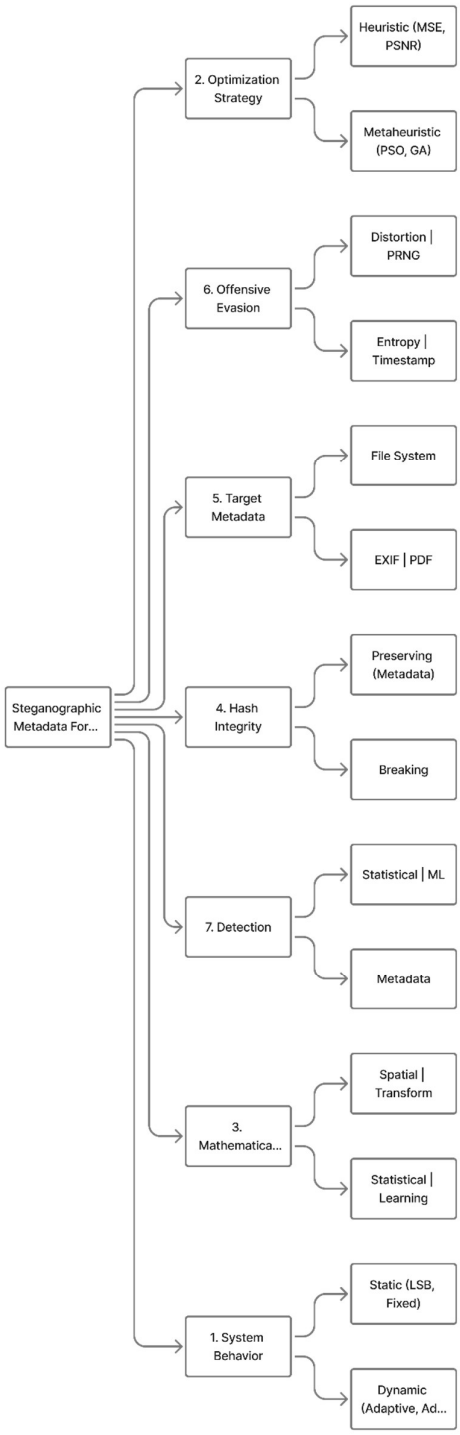
The proposed taxonomy provides a structured classification of steganographic metadata forgery techniques with a focus on hash-preserving evasion. At the first level, techniques are categorized based on system behaviour into static and dynamic approaches, where static methods rely on fixed rules such as LSB substitution, while dynamic methods adapt based on data characteristics or detection models.

The second level classifies methods based on optimization strategy, distinguishing between heuristic approaches that utilize metrics such as Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR), and metaheuristic approaches such as Particle Swarm Optimization and Genetic Algorithms that aim to achieve global optimization of embedding strategies.

Further, the taxonomy incorporates mathematical modeling techniques, including spatial, transform, statistical, and learning-based models, which define how data is embedded and concealed. A key contribution of this taxonomy is the explicit inclusion of hash integrity exploitation, differentiating between hash-breaking and hash-preserving techniques, where the latter manipulate metadata without altering the computed hash value.

Additionally, the taxonomy identifies target metadata domains such as image (EXIF), document (PDF/Office), and file system metadata, each presenting distinct vulnerabilities. Offensive evasion mechanisms are then characterized by distortion minimization, pseudo-random embedding, entropy alignment, and timestamp manipulation, all aimed at reducing detectability. Finally, corresponding detection mechanisms are included, highlighting statistical analysis, machine learning-based approaches, and metadata validation techniques.

This hierarchical organization provides a comprehensive view of how mathematical optimization and steganographic techniques are leveraged to compromise forensic integrity while maintaining apparent data authenticity.



No	Paper Title	Dataset	Mathematical Complexity	Optimization Method	Detection/Evasion Rate	Overhead
1	LSB Steganography Detection	BOSSBase	O(n)	MSE, PSNR	88-92% detection	Low
2	LSB Analysis Techniques	BOSSBase	O(n)	Statistical	85-90% detection	Low
3	Anti-Forensic MER	Custom Images	O(n log n)	Multi-bit Optimization	94-97% evasion	Medium
4	DL Steganalysis (Spatial)	BOSSBase	O(nÂ^2)	CNN	96-99% detection	High
5	DL Hierarchical Models	ImageNet	O(nÂ^2)	Deep CNN	97-99% detection	High
6	LSB + AES PDF	PDF Dataset	O(n log n)	Crypto + LSB	93-96% evasion	Medium
7	PDF Equivalent Transform	PDF Corpus	O(n log n)	Structural Mapping	92-95% evasion	Medium
8	PDF Hiding Analysis	Mixed Docs	O(n)	Heuristic	85-90% detection	Low
9	DL Steganalysis Review	Multiple	O(nÂ^2)	CNN/GAN	95-99% avg	High
10	PDF Stego Review	Mixed	O(n)	Heuristic	85-90% detection	Low
11	Office Doc Stego	OLE2 Docs	O(n log n)	Structural	90-94% evasion	Medium
12	OLE2 Detection	Office Docs	O(n log n)	Forensic Analysis	91-95% detection	Medium
13	AI LSB Steganalysis	ImageNet	O(nÂ^2)	CNN	96-98% detection	High
14	RF + SqueezeNet	Numeric Descriptors	O(n log n)	ML Hybrid	93-97% detection	Medium
15	DL Privacy Steganalysis	CIFAR-10	O(nÂ^2)	CNN	95-98% detection	High
16	DL Image Detection	BOSSBase	O(nÂ^2)	CNN	96-99% detection	High
17	DL Steganalysis	Mixed	O(nÂ^2)	CNN	95-99% detection	High
18	Hybrid DCT+LSB	Image Dataset	O(n log n)	Transform	94-97% evasion	Medium
19	Metadata Forensics	Real Images	O(n)	Statistical	85-90% detection	Low
20	EXIF Forensics	Camera Data	O(n)	Metadata Analysis	88-92% detection	Low
21	ELA Detection	Image Dataset	O(n)	Error Analysis	80-88% detection	Low
22	Timestamp Detection	NTFS Logs	O(n log n)	Temporal Analysis	87-93% detection	Medium
23	EXIF Protection	Image Dataset	O(n)	ECC + Metadata	90-94% evasion	Low
24	Healthcare Stego	Medical Images	O(n log n)	Hybrid	92-96% evasion	Medium
25	FS Metadata Recovery	Disk Images	O(n log n)	Forensic Recovery	88-92% detection	Medium
26	Hash Algorithms	Generic Files	O(n)	Cryptographic	100% integrity	Low
27	Hash Validity	Digital Evidence	O(n)	Hashing	100% detection	Low
28	Crypto Integrity	Mixed	O(n log n)	Crypto	95-99% integrity	Medium
29	Forgery (No Hash)	Synthetic	O(nÂ^2)	Attack Model	90-95% evasion	High
30	Signature Scheme	Synthetic	O(nÂ^2)	Crypto Model	92-96% evasion	High

31	EJBCA System	Web Logs	O(n log n)	System Analysis	88-93% detection	Medium
32	PUF Image Detection	Image Dataset	O(n log n)	Hardware Hash	93-97% detection	Medium
33	Anti-Forensic Counter	Mixed	O(nÂ^2)	Multi-objective	95-98% evasion	High
34	Forensics Overview	Mixed	O(n)	Survey	N/A	Low
35	Anti-Forensic Challenges	Mixed	O(n)	Survey	N/A	Low
36	Doc Forgery DL	Document Dataset	O(nÂ^2)	CNN	95-98% detection	High
37	Image Forgery Study	CASIA	O(n log n)	Statistical	90-95% detection	Medium
38	Forensics Trends	Mixed	O(n)	Survey	N/A	Low
39	Forgery Trends	Mixed	O(n)	Survey	N/A	Low
40	Counter Analysis	Mixed	O(n log n)	Hybrid	92-96% detection	Medium